Practice Questions on GHCR
Dated : 20/2/2026

Q #1:

You are required to build a Docker image from a Dockerfile and publish it to GitHub Container Registry (GHCR), and then pull it on another system.

Task Requirements:
1. Create a simple Dockerfile for a Python Flask application.
2. Build the Docker image locally and tag it appropriately for GitHub Container Registry (ghcr.io).
3. Authenticate Docker with GitHub Container Registry using a Personal Access Token (PAT).

4. Push the Docker image to ghcr.io/<your-github-username>/<image-name>:v1.

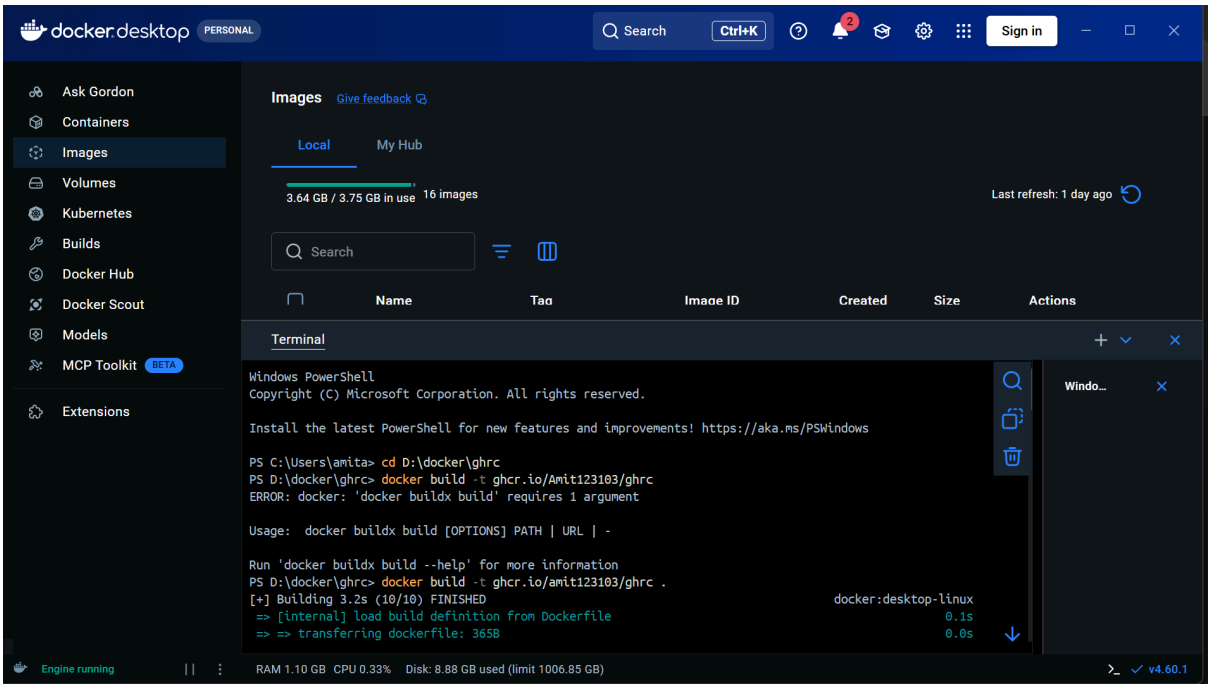
5. Verify that the image is successfully uploaded in your GitHub Packages section.
6. On a different system (or after removing the local image):
• Login again to GHCR.
• Pull the same image.
• Run the container and verify the application is working.

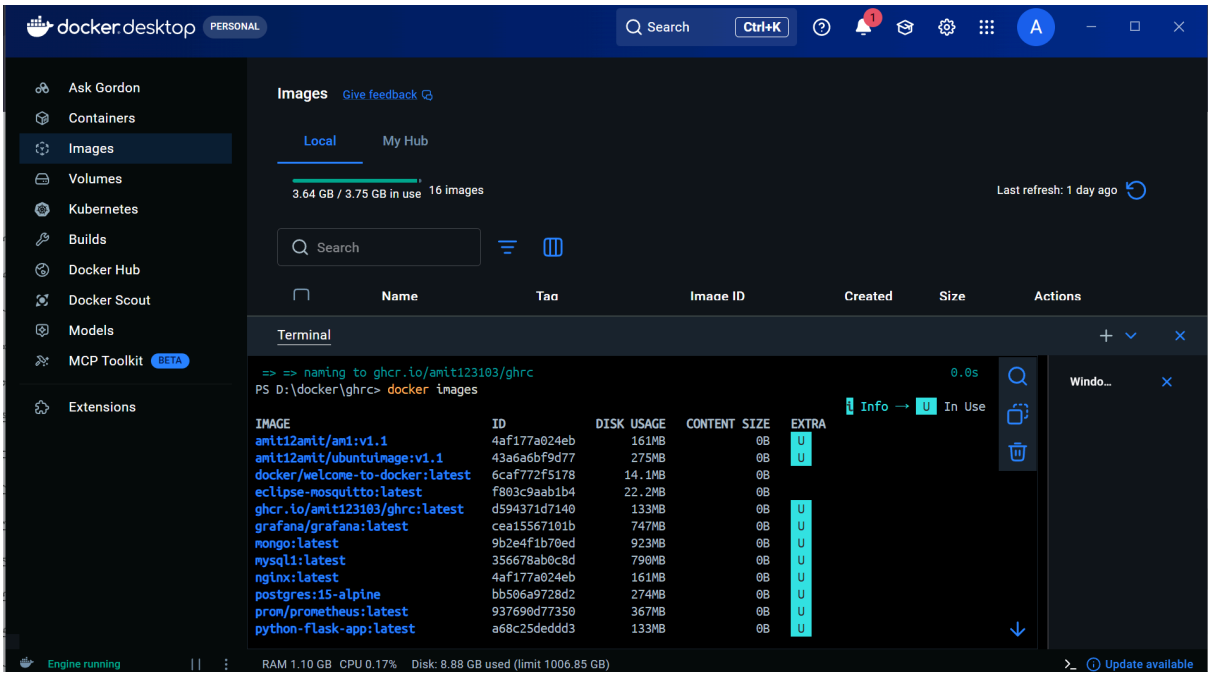
Q#2: You are required to create a customized MySQL Docker image with a preloaded database and publish it to GitHub Container Registry (GHCR), then pull and verify it on another system.

Task Requirements:
1. Create a custom Dockerfile based on the official MySQL image.

2. Configure the Dockerfile to:
• Set root password using environment variables.
• Create a new database (e.g., college_db).
• Import a sample SQL file (init.sql) containing:
• One table (e.g., students)
• At least 3 records.
3. Build the Docker image locally.
4. Tag the image properly for GitHub Container Registry:

Evaluation Rubrics:
• Correct use of MySQL base image
• Proper database initialization
• Correct GHCR tagging format
• Successful push and pull
• Verification of database and table after pulling


docker desktop PERSONAL
Search Ctrl+K
Sign in
Ask Gordon
Containers
Images
Volumes
Kubernetes
Builds
Docker Hub
Docker Scout
Models
MCP Toolkit BETA
Extensions
Images Give feedback
Local My Hub
3.64 GB / 3.75 GB in use 16 images
Last refresh: 1 day ago
Search
Name Tag Image ID Created Size Actions
Terminal
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows
PS C:\Users\amita> cd D:\docker\ghrc
PS D:\docker\ghrc> docker build -t ghcr.io/Amit123103/ghrc
ERROR: docker: 'docker buildx build' requires 1 argument
Usage: docker buildx build [OPTIONS] PATH | URL | -
Run 'docker buildx build --help' for more information
PS D:\docker\ghrc> docker build -t ghcr.io/amit123103/ghrc .
[+] Building 3.2s (10/10) FINISHED docker:desktop-linux
=> [internal] load build definition from Dockerfile 0.1s
=> => transferring dockerfile: 365B 0.0s
Windo...
Engine running
RAM 1.10 GB CPU 0.33% Disk: 8.88 GB used (limit 1006.85 GB)
v4.60.1


docker desktop PERSONAL
Search Ctrl+K
Ask Gordon
Containers
Images
Volumes
Kubernetes
Builds
Docker Hub
Docker Scout
Models
MCP Toolkit BETA
Extensions
Images Give feedback
Local My Hub
3.64 GB / 3.75 GB in use 16 images
Last refresh: 1 day ago
Search
Name Tag Image ID Created Size Actions
Terminal
=> => naming to ghcr.io/amit123103/ghrc 0.0s
PS D:\docker\ghrc> docker images
Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
amit12amit/am1:v1.1 4af177a024eb 161MB 0B U
amit12amit/ubuntuimage:v1.1 43a6a6bf9d77 275MB 0B U
docker/welcome-to-docker:latest 6caf772f5178 14.1MB 0B
eclipse-mosquitto:latest f803c9aab1b4 22.2MB 0B
ghcr.io/amit123103/ghrc:latest d594371d7140 133MB 0B U
grafana/grafana:latest cea15567101b 747MB 0B U
mongo:latest 9b2e4f1b70ed 923MB 0B U
mysql1:latest 356678ab0c8d 790MB 0B U
nginx:latest 4af177a024eb 161MB 0B U
postgres:15-alpine bb506a9728d2 274MB 0B U
prom/prometheus:latest 937690d77350 367MB 0B U
python-flask-app:latest a68c25deddd3 133MB 0B U
Windo...
Engine running
RAM 1.10 GB CPU 0.17% Disk: 8.88 GB used (limit 1006.85 GB)
Update available

Terminal

```
PS D:\docker\ghrc> docker login ghcr.io
Authenticating with existing credentials... [Username: amit123103]

i Info →To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
PS D:\docker\ghrc> docker push ghcr.io/amit123103/ghrc
Using default tag: latest
The push refers to repository [ghcr.io/amit123103/ghrc]
192f851b039e: Layer already exists
a93ac5ea1b61: Layer already exists
3874e5c0429d: Layer already exists
da2c7e522d32: Layer already exists
bb2cb0bd6077: Layer already exists
9f0194fcc39b: Layer already exists
```

RAM 1.10 GB CPU 0.25% Disk: 8.88 GB used (limit 1006.85 GB)

Windo...

Update available

Terminal

```
a8ff6f8cbdfd: Layer already exists
latest: digest: sha256:2d87bc099386b6cd8b15a5b22a4e505da1adfe864b913f401caf1cfc260407fe size: 1991
PS D:\docker\ghrc> docker rmi ghcr.io/amit123103/ghrc
Untagged: ghcr.io/amit123103/ghrc:latest
Untagged: ghcr.io/amit123103/ghrc@sha256:2d87bc099386b6cd8b15a5b22a4e505da1adfe864b913f401caf1cfc260407fe
PS D:\docker\ghrc> docker pull ghcr.io/amit123103/ghrc
Using default tag: latest
latest: Pulling from amit123103/ghrc
Digest: sha256:2d87bc099386b6cd8b15a5b22a4e505da1adfe864b913f401caf1cfc260407fe
Status: Downloaded newer image for ghcr.io/amit123103/ghrc:latest
ghcr.io/amit123103/ghrc:latest
PS D:\docker\ghrc> docker images
                                                                    i Info → U In Use
IMAGE                          ID            DISK USAGE   CONTENT SIZE   EXTRA
amit12amit/am1:v1.1            4af177a024eb        161MB             0B   U
amit12amit/ubuntuimage:v1.1    43a6a6bf9d77        275MB             0B   U
```

RAM 1.09 GB CPU 0.00% Disk: 8.88 GB used (limit 1006.85 GB)

Windo...

Update available

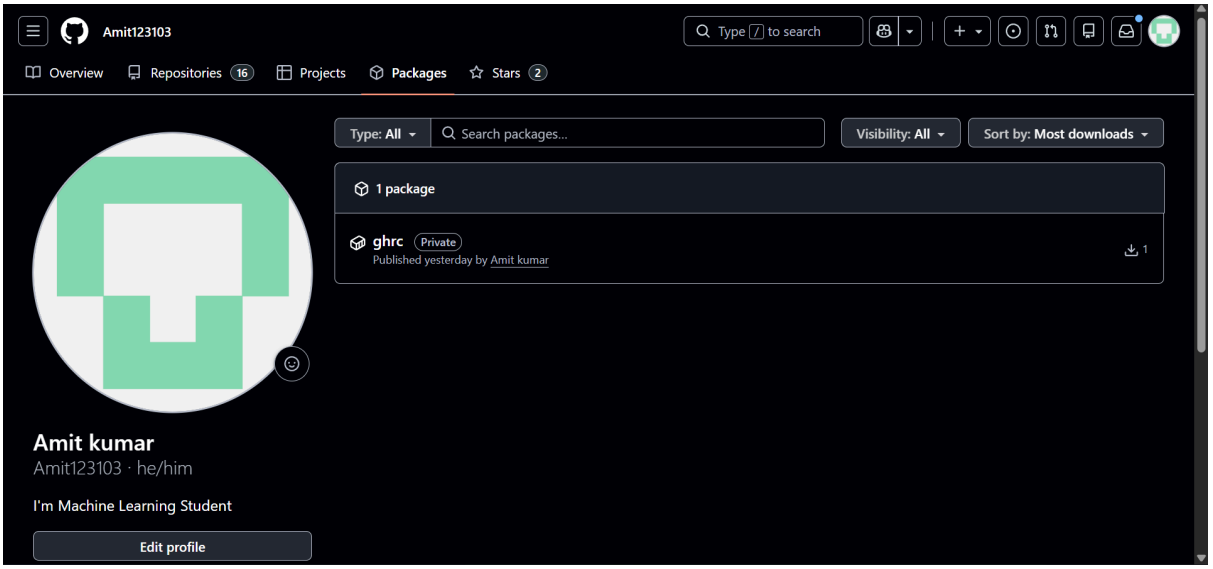
Terminal

```
python-flask-app:latest        a68c25deddd3        133MB             0B   U
python11:latest                d594371d7140        133MB             0B   U
redis:7-alpine                 e08bd8d5a677       41.4MB             0B   U
redis:alpine                   060b91c4b1ee       97.2MB             0B
ubuntu:latest                  493218ed0f40       78.1MB             0B   U
PS D:\docker\ghrc> docker pull ghcr.io/amit123103/ghrc:latest
latest: Pulling from amit123103/ghrc
Digest: sha256:2d87bc099386b6cd8b15a5b22a4e505da1adfe864b913f401caf1cfc260407fe
Status: Image is up to date for ghcr.io/amit123103/ghrc:latest
ghcr.io/amit123103/ghrc:latest
PS D:\docker\ghrc>
```

RAM 1.09 GB CPU 0.08% Disk: 8.88 GB used (limit 1006.85 GB)

Windo...

Update available


Amit123103
Type / to search
Overview
Repositories 16
Projects
Packages
Stars 2
Type: All
Search packages...
Visibility: All
Sort by: Most downloads
1 package
ghrc
Private
Published yesterday by Amit kumar
1
Amit kumar
Amit123103 · he/him
I'm Machine Learning Student
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